F23PP469-VS
VARIABLE SPEED DRIVE
SYSTEM INTERCONNECTION DIAGRAM DRG. No. 2000-EL005-B02-10002

11,000V
RD WH BU
FCF 160A
QAC 160A 400A
BCT1 50 / 1A 2.5VA 5P20
BCT2 50 / 1A 2.5VA 5P20
BCT3 50 / 1A 2.5VA 5P20
SWITCHBOARD EARTH BAR
NOTE 3
SUB FLOOR DROP BOX
BCN 50 / 1A 1.5VA 10P10
BAD
PFV1 RD RD RD
QCE
EARTH BAR
TRANSFORMER EARTH BAR
L1 L2 L3
LINE SIDE INCOMING SUPPLY SHEET 6
X4-1 X4-2 X4-3 X4-4 X4-5 X4-6 X4-7 X4-8 X4-11 X4-12 X4-13
2000-EL003-F16-10011 SHEET 14

OUTGOING MOTOR SUPPLY SHEET 6
T1 T2 T3
CABLE SCREEN EARTHED
M 3~
F23-PP-469
FINAL TAILS PUMP 4
3,300V
800 kW

MOTOR SPACE HEATER CIRCUIT SHEET 6
X2 54 53
F23PP469HT-L
F23PP469HT-N
F23-PP-469
MOTOR HEATER TERMINAL BOX
2000-ME063B-F14-10013
1 (H1) 3 (H2) 0.3kW
ANTI-CONDENSATION HEATER

A80 (PART OF)
RTD TEMPERATURE MONITOR RELAY
SHEET 6
X81
F23-PP-469
MOTOR RTD TERMINAL BOX
2000-ME063B-F14-10013

PT100 RTD INPUT CH1	+ 13 / COM 14 / - 15	F23PP469A-RD / F23PP469A-WH / F23PP469A-BK	MOTOR WINDING RTD (RD PHASE) (TD1)
PT100 RTD INPUT CH2	+ 16 / COM 17 / - 18	F23PP469B-RD / F23PP469B-WH / F23PP469B-BK	MOTOR WINDING RTD (WH PHASE) (TD3)
PT100 RTD INPUT CH3	+ 19 / COM 20 / - 21	F23PP469C-RD / F23PP469C-WH / F23PP469C-BK	MOTOR WINDING RTD (BU PHASE) (TD5)
PT100 RTD INPUT CH4	+ 22 / COM 23 / - 24	F23PP469D-RD / F23PP469D-WH / F23PP469D-BK	MOTOR BEARING RTD (DRIVE END) (BD10)
PT100 RTD INPUT CH5	+ 25 / COM 26 / - 27	F23PP469E-RD / F23PP469E-WH / F23PP469E-BK	MOTOR BEARING RTD (NON DRIVE END) (BD20)
PT100 RTD INPUT CH6	+ 28 / COM 29 / - 30		
PT100 RTD INPUT CH7	+ 31 / COM 32 / - 33		
PT100 RTD INPUT CH8	+ 34 / COM 35 / - 36		

FIELD CABLE TERMINALS

F23-SW-016 11kV FEEDER START/STOP CONTROL 2000-EL003-F16-10011 SHEET 7 3A & 3B
X3 1139 3 → X2 67
X3 3213 17 → X2 68
K41 K41
MV CONTACTOR ENABLE CIRCUIT SHEET 6

F23-SW-016 11kV FEEDER VSD HEALTHY 2000-EL003-F16-10011 SHEET 10 4A
X2 1179 4 → X2 63
X2 1199 25 → X2 64
K41 K41 K41
MV INPUT ENABLE CIRCUIT SHEET 6

F23-SW-016 11kV FEEDER VSD TRIP 2000-EL003-F16-10011 SHEET 10 3A
X2 1219 3 → X2 65
X2 1239 24 → X2 66
K41 K41 K41
MV INPUT TRIP CIRCUIT SHEET 6

F23-SW-016 11kV FEEDER ARC FLASH TRIP 2000-EL003-F16-10011 SHEET 8 5A & 5D
X2 1259 56 → X2 70
X2 1279 44 → X2 69
A30 (PART OF) X2 15 TRIP X2 16
ARC FLASH DETECTION TRIP SIGNAL SHEET 6

F23-DB-101
2000-F23-TLD-EL-10054
RD WH BU BK
R6 W6 B6 25A
415VAC
EARTH BAR
F23DB101-R6 F23DB101-W6 F23DB101-B6 F23DB101-BK6
X4 1 2 3 N PE1
VSD AUXILIARY POWER SUPPLY SHEET 6

F23-DB-201
2000-F23-TLD-EL-10101
RD BK
Q176 10A
240VAC
EARTH BAR
F23DB201-Q176 (RD) F23DB201-Q176 (BK)
X4 5 6 PE2
VSD UPS POWER SUPPLY SHEET 6

TO FINAL TAILS PUMPS
415V SWITCHROOM
F23-COM-001 COMMUNICATION RACK
MODBUS ETHERNET SWITCH
F23-ETH-001
2000-F23-TER-IC-10003
CAT6 — CAT6 — CAT6 — CAT6 — CAT6 — CAT6 — CAT6 — CAT6
X47 ETHERNET ADAPTOR RJ45
SHEET 6

A2 (PART OF) DI-2B 9
X2 13 F23PP469VS-1459
24VDC CONTROL SHEET 6
X2 14 F23PP469VS-1469

SAFE TORQUE OFF CIRCUIT SHEET 6
X86 1 2 3 4
F23PP469VS-1409 F23PP469VS-1419 F23PP469VS-1429 F23PP469VS-1439

F23PP469-LCS
LOCAL CONTROL STATION
X1 1 2 3 4 5 6 E
LOS
START

NOTES

1. WIRE NUMBER FERRULES ON TERMINAL STRIP SHALL BE PREFIXED WITH EQUIPMENT NUMBER e.g. F23PP469-1200, UNLESS OTHERWISE NOTED.
2. EACH WIRE SHALL BE IDENTIFIED AT BOTH ENDS WITH FERRULES.
3. IF THE CABLE SCREENS PASS THROUGH CORE BALANCE CT BEFORE BREAKING OUT THEN THE SCREEN EARTHS MUST BE BOUGHT BACK THROUGH THE CORE BALANCE CT AND TERMINATED TO EARTH.

LEGEND

- ⊗ 11kV SWITCHBOARD TERMINAL
- ⊗ VSD TERMINAL
- ◒ FIELD TERMINAL
- ◑ DCS OR I/O TERMINAL
- ◨ MARSHALLING TERMINAL
- (x) DEVICE TERMINAL

PANEL A06

2000-F23-SCM-EL-10111_Rev1

F23-SW-016 - F23-PP-469 final tails pump 4 VSD feeder - schematic diagram

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